THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS UNIT TEST SET B

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The number of faces of a pentagonal prism is: 1

(a) 8 **(b)** 9 **(c)** 6 **(d)** 7

2. In the Sierpinski triangle, the number of small shaded triangles after stage 2 is: 1

(a) 18 **(b)** 8 **(c)** 9 **(d)** 12

3. A solid has 5 faces and 5 vertices. Using $F + V - E = 2$, the number of edges is: 1

(a) 10 **(b)** 6 **(c)** 8 **(d)** 12

4. A line graph is drawn by joining plotted points with: 1

(a) circles **(b)** sectors **(c)** bars **(d)** line segments

5. In a line graph, a horizontal segment means: 1

(a) no change **(b)** fast increase **(c)** fast decrease **(d)** missing data

6. A bar graph uses: 1

(a) lines only **(b)** circles

(c) bars of heights proportional to values **(d)** dots only

7. If n is any integer, $n^2 + n$ is always: 1

(a) odd **(b)** a square **(c)** prime **(d)** even

8. The sum of three consecutive integers is always divisible by: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 6

9. In a calendar, the dates in a 2×2 block starting at 9 are 9, 10, 16, 17. Their sum is: 1

(a) 50 **(b)** 44 **(c)** 52 **(d)** 54

10. The area of a rectangle of length 12 cm and breadth 7 cm is: 1

(a) 19 cm² **(b)** 38 cm² **(c)** 91 cm² **(d)** 84 cm²

11. The height of a parallelogram of area 72 cm² and base 12 cm is: 1

(a) 6 cm **(b)** 12 cm **(c)** 24 cm **(d)** 60 cm

12. The area of a trapezium with parallel sides 12 cm and 18 cm and height 9 cm is: 1

(a) 216 cm² **(b)** 270 cm² **(c)** 147 cm² **(d)** 135 cm²

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. How many triangles are shaded in stage 3 of the Sierpinski triangle? 2

14. Find the median of 4, 8, 6, 10, 14, 12. 2

15. Write a two-digit number $\overline{ab}$ in algebraic form and its reverse. 2

16. Show that the trick 'Think of a number, add 7, double, subtract 14' always gives double the number. 2

17. A rhombus has diagonals 12 cm and 16 cm. Find its area. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. Find the faces, edges and vertices of a octagonal prism and verify Euler's formula. 3

19. Find the mean and the median of: 22, 24, 26, 28, 30. Which is greater? 3

20. A 2×2 block of a calendar has top-left date 7. Find the four dates, their sum, and show sum $= 4k + 16$. 3

21. A rectangular plot is 25 m by 8 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

22. A Sierpinski triangle starts with one shaded triangle of side 1 unit. After stage 1, find the number of shaded triangles and the side of each. 3

OR

How many edges are there in a Koch curve after stage 3 if stage 0 is one segment? 3

23. The data 100, 120, 140, 160, 180 is given. Compute the mean and median and comment on which is a better measure. 3

OR

The mean of 5 observations is 16. If 4 of them are 12, 15, 18, 20, find the remaining observation. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. (i) What is the cross section of a cube cut diagonally through two opposite edges? 5
(ii) Of a cone cut parallel to the base?

OR

(i) State the steps to draw the Koch snowflake. 5
(ii) If the original side is 9 cm, find the side length and number of sides after stage 2.

25. (i) Show the number trick: 'Think of n, add 5, multiply by 3, subtract 15, divide by 3'. 5
(ii) What does the trick give?
(iii) Create a new trick giving 10 always.

OR

(i) Prove that the sum of the four dates of a 2×2 block of a calendar is $4x + 16$. 5
(ii) Can the sum be 100? 130?

END OF QUESTION PAPER — **Total: 50 Marks**